Mathematical and Computational Methods

Exercise Sheet 9: Integration I: the definite integral and substitution

This sheet accompanies *Unit 9 (Integration I: the definite integral and substitution)*. It exercises Riemann sums and signed area, both parts of the Fundamental Theorem of Calculus, and integration by substitution; no technique from Unit 10 (parts, partial fractions, trigonometric substitution, improper integrals) is needed.

How to use this sheet. Problems are graded by difficulty—$\star$ drill (routine fluency), $\star\star$ core (standard, tutorial level) and $\star\star\star$ challenge (harder or extension)—and organised in three sections: **warm-up drills** for fluency, **tutorial problems** (the core set for the 2-hour class, with the live items flagged ▶) and **further practice** for independent home study.

Solutions. A full worked solution sits after each problem: hidden in the student build, shown in the solutions build. To produce the solutions version, uncomment `\solutionson` in the preamble (or build with `pdflatex "\def\showsolutions{}\input{MCM_Exercises_Unit9}"`) and recompile.

Toolkit. Riemann sums and signed area; the power rule and the table of standard integrals; both parts of the Fundamental Theorem of Calculus (FTC); and integration by substitution, including changing the limits and the $f'/f \to \ln$ pattern.

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1 Warm-up drills

Short, single-idea problems — at least one for every skill in the unit.

Riemann sums and signed area

Exercise 1.1 *($\star$ drill — a right-endpoint Riemann sum)***.** Estimate the area under $f(x) = x^2$ on $[0, 2]$ using $n = 4$ rectangles of equal width and the height at the *right* end of each strip. Is your estimate an over- or under-estimate, and why?

Exercise 1.2 *($\star$ drill — the sign of an integral)***.** Without computing any antiderivative, decide the sign (positive, negative or zero) of each definite integral, and say why.

$$\text{(a)} \int_{-2}^{2} x^3 \, \mathrm{d}x, \qquad \text{(b)} \int_{0}^{1} \mathrm{e}^{-x} \, \mathrm{d}x, \qquad \text{(c)} \int_{\pi}^{2\pi} \sin x \, \mathrm{d}x.$$

Exercise 1.3 *($\star$ drill — sketching the rectangles)***.** A picture first.

(a) Sketch $y = x^2$ on $[0, 2]$ and, on the same axes, draw the four right-endpoint Riemann rectangles of Exercise 1 (width 0.5). Mark whether the rectangles lie above or below the curve.

(b) On separate axes sketch $g(x) = x - 1$ on $[0, 2]$. Shade the region between the graph and the x-axis, marking the part that counts as *negative* signed area. Then compute the net signed area $\int_0^2 (x - 1) \, \mathrm{d}x$ and the total (unsigned) area.

Properties of the definite integral

Exercise 1.4 *(⋆ drill — properties of the definite integral)*. Suppose $\int_0^2 f = 5$ and $\int_2^5 f = -2$. Using only the properties of the definite integral (do *not* find f), evaluate

$$\text{(a)} \int_0^5 f, \quad \text{(b)} \int_5^2 f, \quad \text{(c)} \int_2^2 f, \quad \text{(d)} \int_0^2 3f, \quad \text{(e)} \int_5^0 f.$$

Indefinite integration and the power rule

Exercise 1.5 *(⋆ drill — a polynomial antiderivative)*. Find $\int \left(x^3 - 4x + 2\right) \, dx$.

Exercise 1.6 *(⋆ drill — powers of x first)*. Write each integrand as a power of x first: $\int \left(\sqrt{x} + \frac{1}{x^2}\right) \, dx$.

Exercise 1.7 *(⋆ drill — standard integrals)*. Use the table of standard integrals: $\int \left(2e^x - \frac{3}{x} + 5\cos x\right) \, dx$.

Standard integrals (including arctan/arcsin)

Exercise 1.8 *(⋆ drill — standard forms with a linear inside)*. Evaluate the indefinite integrals

$$\text{(a)} \int e^{3x} \, dx, \quad \text{(b)} \int \sec^2(2x) \, dx, \quad \text{(c)} \int \frac{dx}{9 + x^2}, \quad \text{(d)} \int \frac{dx}{\sqrt{4 - x^2}}, \quad \text{(e)} \int \cos(5x) \, dx.$$

FTC Part 2: evaluating definite integrals

Exercise 1.9 *(⋆ drill — definite integrals by FTC 2)*. Evaluate

$$\text{(a)} \int_1^3 x^2 \, dx, \qquad \text{(b)} \int_0^{\pi/2} \cos x \, dx, \qquad \text{(c)} \int_1^e \frac{dx}{x}, \qquad \text{(d)} \int_0^1 e^{2x} \, dx.$$

FTC Part 1: differentiating an integral

Exercise 1.10 *(⋆ drill — FTC Part 1)*. Differentiate with respect to x (do *not* try to evaluate the integral):

$$\text{(a)} \frac{d}{dx} \int_0^x e^{-t^2} \, dt, \qquad \text{(b)} \frac{d}{dx} \int_2^x \sin(t^3) \, dt, \qquad \text{(c)} \frac{d}{dx} \int_1^x \frac{dt}{1 + t^4}.$$

Substitution (reverse chain rule and f'/f)

Exercise 1.11 *(⋆ drill — substitution)*. Each integrand contains an "inside" function whose derivative is present. Integrate by substitution:

$$\text{(a)} \int 2x \, (x^2 + 1)^5 \, dx, \quad \text{(b)} \int \cos x \, e^{\sin x} \, dx, \quad \text{(c)} \int 3x^2 \, (x^3 + 7)^4 \, dx.$$

Exercise 1.12 *(⋆ drill — the f'/f pattern)*. Spot the f'/f pattern and write down the logarithm:

$$\text{(a)} \int \frac{2x}{x^2 + 5} \, dx, \qquad \text{(b)} \int \frac{\cos x}{\sin x} \, dx, \qquad \text{(c)} \int \frac{e^x}{e^x + 1} \, dx.$$

Exercise 1.13 *(⋆ drill — changing the limits)*. Evaluate $\int_0^1 x \, e^{x^2} \, dx$ by changing the limits to the new variable.

No elementary antiderivative

Exercise 1.14 *(⋆ drill — elementary antiderivatives?)*. For each function, state whether it has an *elementary* antiderivative.

$$\text{(a) } \mathrm{e}^{-x^2}, \quad \text{(b) } x\,\mathrm{e}^{-x^2}, \quad \text{(c) } \frac{\sin x}{x}, \quad \text{(d) } \sin(x^2), \quad \text{(e) } x\cos(x^2).$$

2 Tutorial problems

The core set for the two-hour tutorial. Standard difficulty, anchored on the worked examples of Unit 9, with one problem flavoured for each cohort. Problems flagged ▶ are the live core worked together in the two-hour session; the remaining problems are for completion at home.

Exercise 2.1 *(⋆⋆ core — Riemann sums against the exact area)*. ▶ Let $f(x) = 1 + x^2$ on $[0, 3]$, cut into $n = 3$ strips of width 1.

(a) Compute the right-endpoint and left-endpoint Riemann sums.

(b) Using FTC Part 2, find the exact area and check it lies between your two estimates.

Exercise 2.2 *(⋆⋆ core — simplify, then integrate)*. ▶ Integrate, simplifying first where it helps:

$$\text{(a) } \int \frac{x^2 - 3x + 2}{x}\,\mathrm{d}x, \qquad \text{(b) } \int \left(\mathrm{e}^{2x} + \tfrac{3}{x} - \cos 5x\right)\,\mathrm{d}x.$$

Exercise 2.3 *(⋆⋆ core — FTC evaluation)*. *(FTC evaluation — worked-example anchor.)* ▶ Evaluate $\displaystyle\int_2^4 x^2\,\mathrm{d}x$ and $\displaystyle\int_0^1 \left(3x^2 - 4x + 5\right)\,\mathrm{d}x$, and state in one sentence what the first number represents geometrically.

Exercise 2.4 *(⋆⋆ core — a rate of flow (physics))*. [Physics / applied] Water flows into a tank at the rate $r(t) = 6t$ litres per minute, $0 \le t \le 5$. The volume added between times a and b is $\int_a^b r(t)\,\mathrm{d}t$.

(a) How much water enters during the first 5 minutes?

(b) During which single minute $[t, t+1]$ does the most water enter, and how much?

Exercise 2.5 *(⋆⋆ core — displacement and distance (physics))*. [Physics / applied] A particle moves on a line with velocity $v(t) = 3t^2 - 12t + 9$ ($\mathrm{m\,s^{-1}}$) for $0 \le t \le 3$.

(a) Find the net displacement $\int_0^3 v\,\mathrm{d}t$.

(b) Find the total distance travelled, $\int_0^3 |v|\,\mathrm{d}t$.

Exercise 2.6 *(⋆⋆ core — substitution with a limit change)*. *(Substitution with a limit change — worked-example anchor.)* ▶ Evaluate $\displaystyle\int_0^2 x\sqrt{x^2+1}\,\mathrm{d}x$ by converting the limits to the new variable.

Exercise 2.7 *(⋆⋆ core — a stray x to rewrite)*. Evaluate $\displaystyle\int_0^3 x\sqrt{x+1}\,\mathrm{d}x$. (Here the substitution leaves a stray x behind — rewrite it in terms of u.)

Exercise 2.8 *(⋆⋆ core — two routes to a logarithm)*. ▶ Evaluate two integrals, contrasting the methods:

$$\text{(a) by the } f'/f \text{ pattern, } \int \tan x\,\mathrm{d}x, \qquad \text{(b) by substitution, } \int_1^2 \frac{\ln x}{x}\,\mathrm{d}x.$$

Exercise 2.9 *(⋆⋆ core — completing the square)*. ▶ By completing the square, evaluate $\displaystyle\int \frac{dx}{x^2+2x+2}$.

Exercise 2.10 *(⋆⋆ core — a probability density (data science))*. [Data science / probability] A continuous random variable X has probability density $f(x) = k\,x(1-x)$ on $[0,1]$ and $f(x) = 0$ elsewhere. A density must satisfy $\int_0^1 f = 1$, and $\mathbb{P}(X \le c) = \int_0^c f$ (probability is area).

(a) Find k.

(b) Find $\mathbb{P}\left(X \le \frac{1}{2}\right)$ and comment on the answer.

Exercise 2.11 *(⋆⋆ core — FTC 1 with the chain rule)*. [Mathematics / pure] ▶ Let $g(x) = \displaystyle\int_0^{x^2} \cos t \; dt$. Find $g'(x)$ in two different ways and check they agree.

Exercise 2.12 *(⋆⋆ core — odd and even on a symmetric interval (maths))*. [Mathematics / pure] ▶ Let f be continuous on $[-a,a]$. Using the substitution $u = -x$ on $\int_{-a}^0 f$, prove that $\displaystyle\int_{-a}^{a} f(x)\; dx = 0$ whenever f is *odd*, and that it equals $2\int_0^a f(x)\; dx$ whenever f is *even*.

Exercise 2.13 *(⋆⋆ core — spot the error)*. *(Spot the error / conceptual.)*

(a) A student writes $\displaystyle\int_{-1}^{1} \frac{dx}{x^2} = \left[-\frac{1}{x}\right]_{-1}^{1} = (-1)-(1) = -2$. The integrand is positive everywhere it is defined, so a negative "area" is impossible. What went wrong?

(b) True or false: "$\displaystyle\int_a^b f(x)\; dx$ is a function of x." Justify.

3 Further practice (home)

A larger bank for independent study, running from routine to harder. Challenge and extension problems are marked ⋆⋆⋆.

Indefinite integrals

Exercise 3.1 *(⋆ drill — indefinite drill)*. Find

(a) $\displaystyle\int (4x^3 - 6x^2 + 1)\, dx$, (b) $\displaystyle\int \left(\tfrac{2}{x^3} + 5\sqrt{x}\right) dx$, (c) $\displaystyle\int \left(e^{-x} + \tfrac{1}{1+x^2}\right) dx$, (d) $\displaystyle\int (3\sin x - 4\sec^2 x)\, dx$.

Definite integrals (FTC Part 2)

Exercise 3.2 *(⋆ drill — definite drill)*. Evaluate

$$\text{(a)} \int_1^2 \frac{dx}{x^2},\ \text{(b)} \int_0^{\ln 2} e^x\, dx,\ \text{(c)} \int_0^{\pi/4} \sec^2 x\; dx,\ \text{(d)} \int_{-1}^{1} (x^3+x)\, dx,\ \text{(e)} \int_0^4 \sqrt{x}\; dx.$$

Substitution — indefinite

Exercise 3.3 *(⋆ drill — substitution drill)*. Integrate

$$\text{(a)} \int x^2(x^3+1)^7\, dx,\ \text{(b)} \int \sin^4 x \cos x\; dx,\ \text{(c)} \int \frac{(\ln x)^2}{x}\, dx,\ \text{(d)} \int x\, e^{-x^2}\, dx.$$

Substitution — definite

Exercise 3.4 *(⋆⋆ core — definite substitution)*. Evaluate

$$\text{(a)} \int_1^2 \frac{\ln x}{x}\,\mathrm{d}x, \ \text{(b)} \int_0^{\pi/2} \sin x \cos x\,\mathrm{d}x, \ \text{(c)} \int_0^1 \frac{x}{(1+x^2)^2}\,\mathrm{d}x, \ \text{(d)} \int_0^1 \frac{2x}{x^2+1}\,\mathrm{d}x.$$

The $f'/f \to \ln$ pattern

Exercise 3.5 *(⋆⋆ core — the f'/f pattern)*. Integrate

$$\text{(a)} \int \frac{3x^2+2}{x^3+2x+1}\,\mathrm{d}x, \qquad \text{(b)} \int \frac{\sec^2 x}{\tan x}\,\mathrm{d}x.$$

arctan and arcsin targets

Exercise 3.6 *(⋆⋆ core — inverse-trig standard forms)*. Evaluate the definite integrals

$$\text{(a)} \int_0^{\sqrt{3}} \frac{\mathrm{d}x}{1+x^2}, \qquad \text{(b)} \int_0^1 \frac{\mathrm{d}x}{\sqrt{4-x^2}}, \qquad \text{(c)} \int_0^2 \frac{\mathrm{d}x}{4+x^2}.$$

Substitution leaving an x behind

Exercise 3.7 *(⋆⋆ core — a substitution leaving a stray x)*. Find $\displaystyle\int \frac{x}{\sqrt{x+4}}\,\mathrm{d}x$.

Signed area versus total area

Exercise 3.8 *(⋆⋆ core — signed area and total area)*. For each function find both the net signed area and the total (painted) area over the stated interval.

(a) $\sin x$ on $[0, 2\pi]$.

(b) $x^2 - 1$ on $[0, 2]$.

Integral equations and applications

Exercise 3.9 *(⋆⋆ core — solve for the limit)*. Find the positive value of a for which $\displaystyle\int_0^a (2x+1)\,\mathrm{d}x = 12$.

Exercise 3.10 *(⋆⋆ core — work done on a spring (physics))*. [Physics / applied] A spring obeys Hooke's law with force $F(x) = kx$ ($k = 200\ \mathrm{N\,m^{-1}}$), where x is the extension. The work done stretching it from $x = 0$ to $x = d$ is $W = \int_0^d F(x)\,\mathrm{d}x$. Find the work to stretch the spring by 0.1 m.

Exercise 3.11 *(⋆⋆ core — a triangular density (data science))*. [Data science / probability] X has the triangular density $f(x) = \frac{1}{2}x$ on $[0, 2]$ (and 0 elsewhere).

(a) Verify it is a valid density.

(b) Find $\mathbb{P}(X \le 1)$ as an area.

Challenges and extensions

Exercise 3.12 *(⋆⋆⋆ challenge — FTC 1 with the chain rule)*. Use FTC Part 1 together with the chain rule to differentiate

$$\text{(a)}\ \frac{\mathrm{d}}{\mathrm{d}x}\int_0^{x^2} \mathrm{e}^t\ \mathrm{d}t, \qquad \text{(b)}\ \frac{\mathrm{d}}{\mathrm{d}x}\int_x^{x^2} \ln\left(1+t^2\right)\ \mathrm{d}t.$$

Exercise 3.13 *(⋆⋆⋆ challenge — completing the square, log plus arctan)*. A *linear* numerator over an irreducible quadratic splits into a logarithm plus an arctangent. Evaluate $\int \frac{x+1}{x^2+4x+5}\ \mathrm{d}x$. *(Hint: complete the square below, and write the numerator as a multiple of the denominator's derivative plus a constant.)*

Exercise 3.14 *(⋆⋆⋆ challenge — an odd power under a root)*. Evaluate $\int \frac{x^3}{\sqrt{x^2+1}}\ \mathrm{d}x$. *(Hint: $u = x^2+1$, and write $x^3 = x^2 \cdot x$.)*

Exercise 3.15 *(⋆⋆⋆ challenge — spot the error)*. *(Spot the error.)* A student evaluates $\int_0^{\pi/2} \sin^2 x \cos x\ \mathrm{d}x$ thus:

$$\text{“set } u = \sin x;\ \int_0^{\pi/2} u^2\ \mathrm{d}u = \left[\tfrac{1}{3}u^3\right]_0^{\pi/2} = \tfrac{1}{3}\left(\tfrac{\pi}{2}\right)^3.\text{”}$$

Identify both mistakes and give the correct value.

Exercise 3.16 *(⋆⋆⋆ challenge — no elementary antiderivative (data science))*. [Data science / numerical] The integral $\int_0^1 \mathrm{e}^{-x^2}\ \mathrm{d}x$ has no elementary antiderivative (it gives a value of the error function).

(a) Explain why no FTC 2 “plug-in” will evaluate it exactly.

(b) Bracket its value using left- and right-endpoint Riemann sums with $n = 2$ ($\Delta x = \frac{1}{2}$). Is the true value inside your bracket?